COHORT 16 DSML- MINI PROJECT DOCUMENTATION

Renewable Energy Transition Speed

Date:01/03/2026

DAY 4: ALGORITHM AND ML MODEL TRAINING:

Objective:

Prepare the final feature set and train the Linear Regression model for predicting Renewable Energy Transition Speed.

Part A: Feature Engineering

Work Completed:

- Reviewed the cleaned dataset received from the Data Engineering team.
- Finalized the target variable:
  Transition Speed = Year-over-Year % change in Renewable Energy Share
- Selected relevant predictor variables based on domain knowledge and dataset availability, including: Solar PV price , Energy consumption indicators , GDP/economic indicators , CO_2 emissions (where available)
- Removed non-informative columns such as identifiers and text labels.
- Examined feature distributions to ensure numerical stability.
- Applied StandardScaler to normalize feature magnitudes.
- Verified that no data leakage existed between input features and the target.
- Split the dataset into: Training set (80%) and Testing set (20%)

Outcome:

Feature-ready dataset prepared for model training.

Part B: Algorithm Used

Linear Regression (Final Implemented Model)

Purpose:

- Model the relationship between energy indicators and renewable transition speed.
- Provide interpretable baseline performance.

- Maintain simplicity appropriate for mini-project scope.

Part C: Model Training

Work Completed:

- Imported necessary libraries from scikit-learn.
- Initialized the Linear Regression model.
- Trained the model using the training dataset.
- Generated predictions on the test dataset.
- Verified prediction outputs for consistency.
- Saved the trained model for evaluation.

Outcome:

Linear Regression model successfully trained.

Group Discussion : Day 4

Figure: Team Meeting – 01st March 2026